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PI Name:	Tafrov, Stefan Ph.D.		
Project Title:	DNA Damage Clusters in Human Cell Transformation Induced by Single or Multiple Space Radiation Ion Exposures		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	11973-5000	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
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No. of Post Docs:	3	**No. of PhD Degrees:**	0
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	ED. NOTE (6/27/2011): PI Paul Wilson took over the project from Stefan Tafrov on 5/31/2011, thus changing the end date on this (Tafrov's) grant to 5/30/2011. Tafrov had previously replaced, in December 2009, former PI Betsy Sutherland, who passed away in October 2009. NOTE: Received no cost extension to 2/29/2012, per C. Guidry/JSC (Ed., 6/21/2011)		
Key Personnel Changes/Previous PI:	ED. NOTE (6/27/2011): PI Paul Wilson took over the project from Stefan Tafrov on 5/31/2011. Tafrov had previously replaced, in December 2009, former PI Betsy Sutherland, who passed away in October 2009. SEE SUTHERLAND for reports from FY2007-FY2009 and WILSON for FY2011 reporting.		
COI Name (Institution):			
Grant/Contract No.:	NNJ07HC73I		
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Task Description:	The space radiation environment poses long and short-term risks to human travelers. During a 3-year Mars mission, the nucleus of each cell in an astronaut's body will be hit by ~400 protons and 0.4 high energy, high atomic number (HZE) ions, plus high levels of protons during Solar Particle Events (SPEs). Predicting and mitigating cancer risk from complex radiation fields in outer space is essential for manned exploration of Mars, as well as long-term habitation of the moon. This requires understanding the roles and importance of damages that induce critical changes, cellular repair responses and potentially-oncogenic cellular alterations. Thus, understanding the biological consequences of multiple particle exposures is essential for assessing radiation risks of space travel. Our research focuses on understanding the consequences to human cells of exposure to the levels and kinds of radiations present on the moon and in deep space. We use a human primary cell neoplastic transformation model, in which irradiated human cells acquire growth properties of human cancer cells. We found that exposure to a moderate proton dose followed by a HZE particle gave higher transformation levels than the sum of the two exposures, indicating non-linear effects of multiple radiation exposures. We are continuing our studies of exposing human cells to multiple space ions including protons at low doses, in SPE-simulating proton fields and in combination with HZE ions. Cell survival and transformation levels are measured. Space radiation induces complex DNA damages affecting both DNA strands/double strand breaks (DSBs) and clusters of damaged bases. We are measuring the induction, repair and levels of unrepaired clusters and DSBs, in order to correlate these data to cell killing and transformation. We developed methods for quantifying DNA bistranded clusters in human cells irradiated with low doses of low LET radiation. We showed that Fe ions, X- and X-rays induce clusters in cells with a linear dose-response and that clusters are repaired via repair-intermediate DSBs and abasic clusters. We recently developed a method for measuring DNA cluster levels in cells exposed to relevant occupational levels of low LET radiation, (mSv doses). However, the induction and repair of clusters in cells exposed to simulated complex radiation fields, and the effects of such radiation on induction of potentially oncogenic damage are unknown. We are initiating studies to determine whether levels of slowly-repaired or unrepaired, persistent clusters correlate with the human cell neoplastic transformation frequencies observed with mixed field radiations. We are determining whether conditions or agents that reduce the levels or types of endogenous or radiation-induced clusters also reduce transformation levels, thus suggesting approaches to mitigation of cancer risk during space travel. Melatonin, a human hormone, effectively scavenges radiation-induced hydroxyl radicals that damage DNA. Our preliminary data indicate that melatonin effectively reduces cell transformation by HZE ions. We propose to test its effectiveness in reducing biological damage induced in human cells by single or multiple space particles.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research is directed to revealing the types of DNA damage inflicted by different types of radiation. Also we are studying the intermediate steps in the anchorage-independent growth of cells exposed to radiation. Both approaches could help to develop strategies for better protection of the healthy cells and higher successful rate for the treatment of the tumor cells with radiation.
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Task Progress:	Dr. Chaudhary: The area of each focus formed after exposure to ionizing radiation provides a measure to understand the cellular response to the damage rather than merely the approximate location and extent of a DNA double strand break. Untreated control cells revealed very small sized foci- about 75 % of foci in had an area 0-1 µm2 and fewer than 2 % had area 2-3 µm2 . Almost 57-63% of foci induced by 10, 20 or 40 cGy protons possessed an area in the range of 0-1 µm2. The highest area possessed by any proton induced foci was 4-5µm2 by only less than 1% in 20 and 40 cGy. About 22-33 % of foci had an area of 0-1 µm2 while in case of protons the value was (57-62 %) for same dose of 10, 20 and 40 cGy ions. The exposure of fibroblasts to dual ion beams composed of 20cGy protons followed by 20 cGy iron ions after 30 minutes, induced fewer, less intense and smaller sized foci as compared to those induced by equal doses of ions individually. On reversing the order of beams significantly higher number, more intense and bigger foci were observed. 5. Human Ape1 is important for repair abasic clusters and cell survival (Dr. Das, Bennett): Dr. Das: Cluster damages in DNA-two or more closely spaced oxidized bases, strand breaks as well as abasic sites are induced by high and low-Linear Energy Transfer radiation and certain radiomimetic drugs and have been found to be repair resistant and potentially mutagenic. In the present study we found that abasic cluster damages are also induced by low-Linear Energy Transfer radiation (low LET) even at very low doses of X-irradiation in primary human Fibroblast cells. X-irradiated primary fibroblasts showed dose dependent upregulation of the DNA repair enzyme, Human abasic endonuclease (hApeI) at low doses of X-ray in the range of 10-100 cGy in response to the induction of the abasic sites. We found that the abasic clusters in primary fibroblasts are more lethal to the cells when the hApeI enzyme expression is downregulated by transfecting the primary fibroblasts with hApeI siRNA as determined by clonogenic survival assay. Endonuclease activity of hApeI was found to be directly proportional to hApeI gene silencing efficiency. DNA repair profile shows that processing of the abasic clusters is delayed in hApeI siRNA silenced fibroblasts, which challenges the survival of the cells even at very low doses of X-rays. Thus, the present study is the first attempt to understand the induction of Cluster DNA damage at very low doses of Low-LET in primary human fibroblasts and their processing by DNA repair enzyme ApeI and their relation with the survival of the cells. The amount of ApeI enzyme in a given number of cells is determined by quantitative western blot. The hApeI level in the siRNA transfected fibroblasts is found to decrease with time, reaching a minimum after 48h hours of transfection and again increases nearly to normal after 96 h. 6. Yields of clustered DNA damage induced by charged particle radiations of similar kinetic energy per nucleon: LET-dependence in different DNA microenvironments (Dr. Keszenman). Dr. Keszenman: To study the LET dependence of the biological effects of densely ionizing radiation when the LET modification is related to changes in the ionization density along the track, we have determined the types, yields and spectrum of clustered DNA damages induced by charged particle beams of different Z and similar kinetic energy per nucleon in different DNA microenvironments. Using a megabasepair DNA biological model we quantified the frequencies of DNA double strand breaks, oxypurine and abasic sites clusters induced by ionizing radiation through pulsed field electrophoresis and number average lengths analysis. Yeast chromosomal DNA embedded in agarose plugs in solutions of different free radical scavenging capacity was irradiated with charged particle beams of 1 GeV/nucleon protons (Z = 1), 1 GeV/nucleon oxygen (Z = 8), 980 MeV/nucleon titanium (Z = 22) and 968 MeV/nucleon iron (Z = 26). Results show that the yields of total DNA clustered damages calculated per absorbed dose in Gy induced by charged particle radiation in 10 mM phosphate buffer (non-radioquenching) decreased with increasing LET, meanwhile, in 10 mM Tris (radioquenching) exhibited minor variations. However, the total damage yields calculated per particle fluence increased with LET in both conditions, indicating a higher efficiency per particle to induce DNA clustered damage. 7. System for in vivo studying DNA repair of complex damages (Dr. Moscariello). Dr. Moscariello: DNA damaging agents can induce clustered lesions or multiple damaged sites (MDSs) on the same or opposite DNA strands. Attempt to repair MDS can generate closely opposed single strand break intermediates that can convert non-lethal or mutagenic base damage into double strand break (DSB). We wondered the biological relevance of such lesions to generate DSB by using a genetic approach in yeast S. cerevisiae. We constructed a diploid strain with a chromosomal context targeted by integrative fragments with different MDSs mapping in a region of shared homology between the chromosomes. Following the integration whether the repair of the MDS generates DSBs the predicted outcomes of the Homologous Recombination repair pathway (HR) will be detected since the engineered yeast strain has a hemizygotic distribution of the selectable markers LEU2 and HIS3 on the chromosome couple XV. To facilitate the synthesis of integrative fragments with MDS the vector pMM-25 was constructed. Combinations of MDSs can be generated by ligation-based replacement of the central portion of the integrative fragment with commercial oligonucleotides housing (or carrying) damaged bases. Since it is possible the synthesis of an extremely broad set of complex MDS our system is fit to study the effect of complex DNA lesions induced by physical and chemical agents.
Bibliography Type:	Description: (Last Updated: 03/11/2021)
Articles in Peer-reviewed Journals	Keszenman DJ, Sutherland BM. "Yields of clustered DNA damage induced by charged particle radiations of similar kinetic energy per nucleon: LET-dependence in different DNA microenvironments." Radiation Research. 2010 Aug 4;174(2):238-50. http://www.rrjournal.org/doi/abs/10.1667/RR2093.1 , Aug-2010